Audit of the ICLR workshop figure set and supporting analyses

Repository: `ukc10014/steering_across_personas`
Audited branch: `workshop-figures`
Original audit snapshot: `2137a7ea84b05a3e996ea976d57e4e3dc2b38088`
Post-audit fixes reviewed: `157921de5961c079e5e74085b994e86ef9e00065`
Audit date: 2026-09-03

Executive summary

I do **not** see evidence that the coding agent has made a catastrophic mathematical or statistical error in the core analyses. In fact, the strongest parts of the current pipeline are unusually careful: the quadratic geometry statistics are cross-fitted; the question bootstrap is paired across arms; the vector ANOVA reconstructs exactly and checks its base-invariance algebra at runtime; the global-map analysis is genuinely cross-validated; and the new A/B logit extraction reads the correct next-token position without a judge or sampling.

I do, however, find several things that should be fixed before the workshop paper is written around the figures:

1. **The `iclr2026/figures/` set is obsolete.** It was hand-reconstructed from numbers in prose/chat and is explicitly marked superseded. The authoritative current figures are in `workshop_iclr/`. The current `iclr2026/main.tex` still uses the old figure set.
2. **Appendix Figure A2 had a real units/label bug, now corrected in `157921d`.** The plotted per-cell C×T×P quantity is a cross-fitted **squared energy** $\langle CTP^A, CTP^B\rangle$, not a linear magnitude. The corrected presentation now gives both energy and its square-root linear magnitude.
3. **The old "rotation with dose implies constitution content" interpretation was stale and is now superseded in `157921d`.** Appendix Figure A4 shows that `random_perm` exhibits essentially the same dose-dependent rotation. The surviving claim is about the **destination** of the shift, not the fact that it rotates.
4. **Figure 3's measured dose ladders are strong; its bottom "corrected to dose 1" panels are model-based.** Those panels locally extrapolate/interpolate using estimated slopes (and borrow slopes for arms with only one point) without propagating uncertainty in dose or slope. They are useful summaries but should not carry fine-grained significance claims until a joint dose–outcome bootstrap is done.
5. **Figure 4 is now a statistically resolved trained-vs-untrained directional result.** Commit `157921d` adds pairwise bootstrap intervals: the weakest trained×trained pair is 0.465 [0.447, 0.482], while the strongest trained×untrained pair is 0.292 [0.263, 0.321]. The result supports a shared **OCT-training-associated destination direction**, not yet a

semantic-content direction; a sham-trained LoRA is still needed to separate training-process effects from meaningful character signal.

6. **Figure 5 passed the requested affine/local-near-zero robustness checks in `157921d`.** Base indifference is populated, trained-arm residual curvature is small, and a model-free local estimate near indifference preserves the target contrast. The forced-choice result is therefore substantially more secure, though it remains a forced-choice rather than free-form behavioral result.
7. **A preregistration does exist, but its scope matters.** `prereg/2026-07-17-v1.md` registers `impulsivity` for the impulsiveness constitution. The later `impulsivity + risk_taking` pair was fixed before the logit run but is post hoc relative to the preregistration; the registered single-trait result should carry the confirmatory claim.
8. **Raw activation and A/B-logit cell files are not versioned in GitHub.** I can trace the code and committed summary JSON/figure CSVs, but cannot independently recompute every number from raw samples from this repository snapshot alone. For a strong reproducibility package, archive the `_qcache` and/or raw cell outputs externally with hashes.

My overall confidence by figure:

Figure	Core computation	Statistical interpretation	Main action
Fig 0 schematic	N/A	**Stale**	update pipeline labels / add separate Fig 5 branch
Fig 1 decomposition	**Strong**	**Mostly strong**	soften "this is CT resolved into traits"
Fig 2 C×T×P	**Strong**	**Strong with scope caveats**	fix "preregistered" history; fix per-cell energy units elsewhere
Fig 3 dose + controls	**Strong for measured ladders**	**Moderate for dose-1 corrections**	joint x/y bootstrap; soften "inside range" wording
Fig 4 shared direction	**Strong estimator**	**Strong trained-vs-untrained separation; semantic attribution incomplete**	sham control

Figure	Core computation	Statistical interpretation	Main action
Fig 5 A/B preference	**Strong extraction**	**Promising; needs affine robustness**	local-near-zero/nonlinearity check
Fig A1 calibration	**Strong**	**Strong calibration**	keep "~700×" approximate
Fig A2 diagnostics	**Strong except label**	**Mostly strong**	**fix squared-energy axis**
Fig A3 L20	**Strong**	**Strong replication**	same dose-correction caveat as Fig 3
Fig A4 rotation control	**Strong negative control**	**Qualitative, not formal curve fit**	update stale §3.3 claims
Fig A5 signed validity	**Sound diagnostic**	**Correct negative conclusion, wording can soften**	say "not validated as semantic sign"

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1. Which directory is actually authoritative?

The user-facing LaTeX scaffold is in `iclr2026/`, but the current figures are **not**.

`iclr2026/FIGURE_NOTES.md` explicitly begins:

```
SUPERSEDED (see workshop_iclr/)
```

and explains that the old figures were reconstructed by hand from values quoted in prose. `workshop_iclr/README.md` likewise says not to mix the two sets.

Therefore this audit treats:

- **authoritative figure package:** `workshop_iclr/`
- **current detailed results write-up:** `docs/results/llama31_8b_extraction_and_geometry.md`
- **stale paper scaffold to be updated:** `iclr2026/main.tex`

The authoritative package contains 11 figures:

1. `fig0_schematic`
2. `fig1_decomposition`
3. `fig2_ctp`
4. `fig3_dose_and_control`
5. `fig4_shared_direction`

6. `fig5_behavioral_preference`
7. `figA1_dose_calibration`
8. `figA2_diagnostics`
9. `figA3_layer20`
10. `figA4_rotation_control`
11. `figA5_signed_validity`

For each numerical figure there is an auditable CSV in `workshop_iclr/data/`, written by the plotting script itself.

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2. Common data pipeline

Understanding the figures is easier if the shared lower-level pipeline is made explicit.

2.1 CAA activation extraction

Raw extraction code: `pipeline/2c_caa_activations.py`

Key functions: - `format_caa_user_message(...)` - `get_answer_letter(...)` - `find_answer_token_position(...)` - `extract_caa_activations(...)`

For constitution/intervention c, trait t, persona p, question q, and direction $+$ or $-$, let the hidden state at the answer-letter token be

$$h^{+}_{c,t,p,q}, \qquad h^{-}_{c,t,p,q} \in \mathbb{R}^{4096}.$$

The CAA trait vector is

$$V_{c,t,p} = \frac{1}{N_t} \sum_{q=1}^{N_t} \left(h^{+}_{c,t,p,q} - h^{-}_{c,t,p,q} \right).$$

Here: - $N_t \approx 500$ questions per trait (empathy has 499), - p ranges over 10 semantic personas for the headline analyses, - `null` and `nonsense` also exist as extraction controls, - layers 15 and 20 are cached.

Raw files are:

`outputs/<arm>/caa_activations/{persona}_{trait}_{pos|neg}.pt`

These bulk files are **not committed to GitHub**.

Attention-mask history

The historical activation archive uses the legacy attention-mask path. A bug was discovered: `pad_token_id` equals `<|eot_id|>` on Llama-3.1, so the old token-identity mask hid genuine turn-boundary tokens. The diagnostic in the results document says the bug is real but the measured impact is small enough that the archived geometry remains usable; all arms being compared use the same legacy extraction. This should remain in limitations/reproducibility because dispersion was among the somewhat more mask-sensitive statistics.

2.2 Per-question cache

Producer: `scripts/build_question_cache.py`

Output:

`outputs/_qcache/{arm}_L15_20.npz`

Layout:

$$(\text{trait}, \text{persona}, \{+, -\}, \text{question}, \text{layer}, \text{hidden}).$$

The cache stores padded question slots but also `n_questions_per_trait`; downstream code slices to the real count, so empathy's 499 questions do not accidentally resample a fake padded question.

The cache itself is also not committed.

2.3 Cross-fitting used by the geometry analyses

Library: `scripts/geometry_lib.py`

Important functions: - `prepare_diff(...)` - `batch_vectors(...)` - `half_splits(...)` - `bootstrap_half_splits(...)` - `dispersion_crossfit(...)` - `rdm_crossfit(...)` - `procrustes_cv(...)` - `linear_map_cv(...)`

For a noisy estimated vector V, direct squared norms are positively biased:

$$\mathbb{E}\|\hat{V}\|^2 = \|V\|^2 + \mathbb{E}\|\varepsilon\|^2.$$

The code instead constructs independent estimates from disjoint question halves A, B and uses

$$\mathbb{E}\,\langle \hat{V}^A, \hat{V}^B \rangle = \|V\|^2$$

when the half-sample errors are independent and mean-zero.

The bootstrap implementation is particularly important: duplicated copies of a resampled question are kept on the **same** side of the A/B split. That prevents a bootstrap-resampled question from leaking into both halves and destroying the independence on which the cross-fit estimator depends.

This is one of the strongest parts of the analysis.

2.4 Random LoRA controls

Constructor: `scripts/make_random_lora.py`

Three controls:

IID

Reuse or redraw A, draw Gaussian B, then rescale B so each module satisfies

$$\|\Delta W_{\text{random}}\|_F = \|\Delta W_{\text{ref}}\|_F.$$

Spectrum-matched

Construct

$$\Delta W_{\text{spec}} = U \operatorname{diag}(s_{\text{ref}})\, V^\top$$

with random orthonormal U, V, preserving the real update's singular values but randomizing directions.

Per-module permutation

Construct

$$\Delta W_{\text{perm}} = P_{\text{out}}\, \Delta W_{\text{ref}}\, P_{\text{in}},$$

with independent coordinate permutations per module. This preserves Frobenius norm, singular values, and singular-vector entry distributions while destroying alignment to the model's actual coordinates and cross-module correspondence.

These are good controls, but at matched *functional* dose they must be scaled to $s \approx 16$–19, much larger in weight space than the trained arms.

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3. Figure-by-figure audit

Figure 0 — `fig0_schematic`

Plot script: `workshop_iclr/scripts/fig0_schematic.py`
Data source: none; schematic only.

What it is intended to say

The diagram shows:

OCT/random intervention → CAA extraction → $V_{c,t,p}$ → geometry/decomposition statistics.

Audit

Verdict: RED for current labeling, not for science.

There are at least three stale elements:

1. It says **"4 constitutions × 4 dose rungs."** Only `goodness`, `impulsiveness`, and `misalignment` have four-rung dose ladders. `mathematical` has one measured point.
2. Its statistics box associates "Fig 5 sign" with dG. Current Figure 5 is no longer a geometric sign; it is a **separate forward-pass A/B logit experiment**.
3. Figure 5 does not descend from the same cached $V_{c,t,p}$ array. It needs its own branch from the model/CAA prompts to answer-letter logits.

Recommendation

Redraw the schematic as two downstream branches:

```
CAA prompts
   +- answer-token activations -> V_c,t,p -> Figs 1-4, A2-A5
   `- pre-answer logits -> signed A/B preference -> Fig 5
```

That would make the paper's central methodological distinction much clearer.

Figure 1 — `fig1_decomposition`

Plot script: `workshop_iclr/scripts/fig1_decomposition.py`
Figure data: `workshop_iclr/data/fig1_decomposition.csv`
Analysis JSON: `outputs/analysis/common_shift.json`
Analysis producer: `scripts/common_shift.py`

Important analysis functions: - `make_preps(...)` - `stats_for_trait(...)` - `main(...)`

Data path

```
caa_activations/*.pt
  → scripts/build_question_cache.py
  → outputs/_qcache/{arm}_L15_20.npz
  → scripts/common_shift.py
  → outputs/analysis/common_shift.json
  → workshop_iclr/scripts/fig1_decomposition.py
  → workshop_iclr/data/fig1_decomposition.csv
  → fig1_decomposition.pdf
```

Mathematics

Define adapter-induced movement:

$$dV_{c,t,p} = V_{c,t,p} - V_{\mathrm{base},t,p}.$$

Persona-common shift:

$$dG_{c,t} = \frac{1}{P}\sum_p dV_{c,t,p}.$$

For independent question halves A, B, the code estimates

$$\widehat{\|dG\|^2} = \mathbb{E}_s\left[dG^A \cdot dG^B\right],$$

and total mean squared movement

$$E_p\widehat{\|dV_p\|^2} = \mathbb{E}_s\left[\frac{1}{P}\sum_p dV_p^A \cdot dV_p^B\right].$$

Panel A: persona-common share

$$S_{c,t} = \frac{\widehat{\|dG_{c,t}\|^2}}{E_p\widehat{\|dV_{c,t,p}\|^2}}.$$

This is legitimately a **share of squared movement** because, at population level,

$$\frac{1}{P}\sum_p \|dV_p\|^2 = \|dG\|^2 + \frac{1}{P}\sum_p \|dV_p - dG\|^2.$$

Panel B: common-shift magnitude

Base scale:

$$b_t = \mathrm{mean}_p \, \|V_{t,p}^{\mathrm{base}}\|,$$

estimated cross-fitted.

The plotted quantity is

$$M_{c,t} = \frac{\sqrt{\widehat{\|dG_{c,t}\|^2}}}{b_t}.$$

So $M = 1$ means that the persona-common shift is about one base trait-vector norm long.

Does the implementation match the formula?

Yes, substantially.

`stats_for_trait(...)` uses the same question halves across arms, giving a paired comparison. Bootstrap CIs are question-resampling CIs, not population CIs over traits/personas.

One minor technical note: the base norm averages a square root of the cross-half dot product over splits. Because square root is concave, there can in principle be a small downward Jensen bias. With ~500 questions and high vector reliability this is unlikely to materially affect the 1.72× selectivity claim, but it is worth knowing.

What it currently shows

At L15, the `impulsiveness` common-shift magnitudes are approximately:

- risk-taking: 1.097
- impulsivity: 1.154
- other six: mostly ~0.62–0.70

giving roughly

$$\frac{\text{mean target}}{\text{mean other}} \approx 1.72.$$

The common-shift squared share is large across all trained arms, typically about 0.61–0.84 by trait.

Narrative I agree with

- Much of OCT-induced movement is persona-common.
- `impulsiveness` is unusually **selective in magnitude** for risk-taking and impulsivity.
- This is not a signed "more impulsive" result.

Narrative I would change

The workshop README says Panel B is "Figure 2's C×T term resolved into traits." That is too literal.

$M_{c,t}$ is the raw common-shift magnitude for a constitution × trait cell. It contains any persona-common contribution of μ, C, T, and CT. The **pattern of constitution-specific deviations across traits** is related to CT, but the raw Panel B values are not simply the CT tensor.

Safer wording:

> "Figure 1 resolves the persona-common intervention by trait; the constitution-dependent heterogeneity across traits is the phenomenon summarized by the C×T term in Figure 2."

Graph suggestion

Keep this in the main paper. Once the second impulsiveness seed lands, either: - show the two seeds separately with a mean, or - show seed replication in a small inset.

Figure 2 — `fig2_ctp`

Plot script: `workshop_iclr/scripts/fig2_ctp.py`
Figure data: `workshop_iclr/data/fig2_ctp.csv`
Analysis JSON: `outputs/analysis/three_way_interaction.json`
Producer: `scripts/caa_three_way_interaction.py`

Important functions: - `anova_components(X)` - `term_ss(compA, compB, shape)` - `build_tensors(...)` - `crossfit_bands(...)` - `check_partition(...)` - `df_shares(...)`

Data path

```
caa_activations
 → _qcache
 → caa_three_way_interaction.py
 → outputs/analysis/three_way_interaction.json
```

→ `fig2_ctp.py`
→ `fig2_ctp.csv`
→ `fig2_ctp.pdf`

Mathematics

The code first normalizes each trait:

$$X_{c,t,p} = \frac{V_{c,t,p} - V_{t,p}^{\text{base}}}{s_t},$$

where

$$s_t = \text{mean}_p \, \|V_{t,p}^{\text{base}}\|$$

is cross-fitted once for the point analysis and then **held fixed** through the bootstrap.

It uses the exact balanced three-way decomposition:

$$X = \mu + C + T + P + CT + CP + TP + CTP.$$

For example,

$$CTP_{c,t,p} = X_{c,t,p} - X_{c,t,\cdot} - X_{c,\cdot,p} - X_{\cdot,t,p} + X_{c,\cdot,\cdot} + X_{\cdot,t,\cdot} + X_{\cdot,\cdot,p} - X_{\cdot,\cdot,\cdot}.$$

The code checks numerically that summing all eight broadcast terms reconstructs X.

Cross-fitted term energy

For each term K,

$$SS_K = \sum_{c,t,p,h} K_{c,t,p,h}^{A} K_{c,t,p,h}^{B}.$$

The plotted partition share is

$$f_K = \frac{SS_K}{SS_{\text{total}}},$$

where

$$SS_{\text{total}} = \sum_{c,t,p,h} X_{c,t,p,h}^{A} X_{c,t,p,h}^{B}.$$

Base subtraction

For terms involving c, subtraction of $V_{\mathrm{base,t,p}}$ is algebraically annihilated by the constitution contrasts. The code verifies numerically:

$$CTP(V - V_{\mathrm{base}}) = CTP(V),$$

and similarly for C, CT, CP.

I agree with this.

Point result, trained L15

Current source CSV:

T	.3720
TP	.1749
CT	.1700
μ	.1241
C	.0666
P	.0428
CTP	.0363
CP	.0133

Terms with no constitution index sum to:

$$.3720 + .1749 + .1241 + .0428 = .7138.$$

So roughly 71.4% of the normalized squared change has no constitution index at all.

Also:

$$CT/CP \approx .1700/.0133 \approx 12.8.$$

This supports “constitutional differences are far more trait-dependent than persona-dependent.”

Matched-DF control

Because CTP degrees of freedom depend on the number of constitution levels, the code compares the 3-arm untrained band against **all four 3-arm subsets** of the 4-arm trained band.

L15 trained 3-arm subsets:

.0288 to .0347

versus untrained:

.0724.

L20 trained subsets:

.0354 to .0454

versus untrained:

.1070.

This is the correct comparison.

Statistical audit

Verdict: GREEN, with scope caveats.

Strengths: - exact factorial algebra; - runtime reconstruction check; - cross-fit quadratic terms; - same question splits across bands; - matched number of constitution levels; - both share and absolute RMS reported; - question bootstrap; - explicit recognition that question sampling noise is highly structured rather than independent per cell.

Scope: - CIs condition on these 4 constitutions, 10 personas, and 8 traits; - they do not establish a population theorem about "constitutional training in general."

Important terminology issue

The script/docs repeatedly call this "the prereg's question." Elsewhere in the repository, the older character-arms document explicitly says `PREREG.md` was never written, and I do not find one in the repository.

Unless there is an external timestamped preregistration, use:

> "the original planned analysis question"

or

> "the registered design note"

rather than "preregistered."

Narrative I agree with

> The strong hypothesis that character training rewrites trait representations in a highly persona-specific manner is not supported. Most structure is lower-order, and the CTP share is small.

Narrative I would qualify

Do **not** write:

> "Training suppresses persona-specific interaction."

The random arms are close to the measured coherence cliff; model damage could itself create idiosyncratic cells. The random comparison only establishes that **nonzero CTP is not evidence of semantics**.

Figure 3 — `fig3_dose_and_control`

Plot script: `workshop_iclr/scripts/fig3_dose_and_control.py`
Shared builder: `workshop_iclr/scripts/dose_panels.py`
Figure data: `workshop_iclr/data/fig3_dose_and_control.csv`

Sources: - `outputs/llama-3.1-8b-goodness/analysis/geometry_L15.json` - `outputs/analysis/functional_dose_with_random.json`

Producers: - `scripts/geometry_analysis.py` - `scripts/geometry_lib.py` - `scripts/functional_dose.py` - random-arm siting also uses `scripts/activation_dose_probe.py`

Functional-dose formula

The primary x-axis is answer-token hidden-state displacement:

$$\delta_a = \frac{\mathrm{mean}_{q,\mathrm{cells}} \|h_a - h_{\mathrm{base}}\|_2}{\mathrm{mean}_{q,\mathrm{cells}} \|h_{\mathrm{base}}\|_2}.$$

The figure normalizes this by `goodness` at $s = 1$:

$$d_a = \delta_a / \delta_{\mathrm{goodness,s=1}}.$$

This is deliberately a **functional** rather than weight-norm dose.

RDM preservation

For persona pair p, q within trait t,

$$d^2_{a,t,pq} = \mathbb{E}_s \left[(V^A_{a,t,p} - V^A_{a,t,q}) \cdot (V^B_{a,t,p} - V^B_{a,t,q}) \right].$$

With ten personas there are 45 pairwise distances per trait.

The dose figure uses the **raw aggregate Spearman** between the arm RDM and base RDM:

$$R_a = \mathrm{mean}_t\, \rho_S \left(\mathrm{RDM}_{a,t}, \mathrm{RDM}_{\mathrm{base,t}}\right).$$

This is intentionally not the attenuation-corrected RDM statistic quoted in an earlier section of the long results document. The two should not be mixed numerically.

Persona dispersion

Within one trait,

$$D_{a,t} = \frac{1}{P} \sum_p \|V_{a,t,p} - \bar{V}_{a,t}\|^2.$$

The linear spread ratio used in the figure is effectively

$$RMS_a = \sqrt{\mathrm{mean}_t\, \frac{D_{a,t}}{D_{\mathrm{base,t}}}}.$$

This is a linear-radius interpretation of a squared dispersion statistic.

Top dose-response panels

These use **measured dose rungs**, not an interpolated common grid.

That is important and good.

RDM

The within-adapter ladder establishes causally that RDM preservation falls as a given trained LoRA is scaled up.

`goodness` and `impulsiveness` lie close to the same dose-response curve; `misalignment` is lower.

Dispersion

The three arms do not lie on one curve. `impulsiveness` has a much flatter response and crosses the other two.

I agree with the conclusion that "geometric change" is not one scalar phenomenon.

Bottom matched-dose panels

This part is more model-dependent.

For a ladder arm, `dose_panels.py` takes the two highest-dose measured rungs and estimates:

$$m_a = \frac{y_{\text{hi}} - y_{\text{lo}}}{d_{\text{hi}} - d_{\text{lo}}}.$$

Then:

$$\hat{y}_a(d = 1) = y_a + m_a(1 - d_a).$$

For arms without their own ladder: - `mathematical` borrows the mean trained slope; - `random_iid` and `random_spec` borrow the `random_perm` slope.

The plotted CIs are shifted by the same deterministic correction. They do **not** propagate: - uncertainty in measured dose, - uncertainty in the local slope, - uncertainty from borrowing a slope from another arm.

Current corrected L15 values

RDM: - goodness .8335 - mathematical .8783 - impulsiveness .8264 - misalignment .7677 - random_iid .8857 - random_spec .8461 - random_perm .7609

Dispersion RMS: - goodness .6969 - mathematical .7916 - impulsiveness .7980 - misalignment .6080 - random_iid .7163 - random_spec .7307 - random_perm .8026

Audit

Top measured ladder panels: GREEN.

Bottom dose-1 correction panels: YELLOW.

They are reasonable descriptive normalization, but not clean experimental observations.

Concrete wording correction

The current Figure 3 docstring says untrained arms “land inside the trained range” on RDM. Strictly, at L15:

$$0.8857_{\text{random iid}} > 0.8783_{\text{mathematical}},$$

so one untrained value is slightly above the trained maximum. The intervals overlap, so the scientific story is unchanged.

Use:

> "The untrained controls overlap and slightly extend the trained RDM range."

Another wording correction

The code says that because RDM and dispersion pick different anomalous arms, "neither is one idiosyncratic adapter." With one trained seed per constitution, that is too strong: different adapters can be idiosyncratic in different ways.

Use:

> "No single scalar dose law explains both outcomes; adapter-specific or content-specific mechanisms remain possible."

Recommended next analysis

A **joint dose–outcome bootstrap** would turn the bottom panels from useful interpolation into a much cleaner inferential figure.

Figure 4 — `fig4_shared_direction`

Plot script: `workshop_iclr/scripts/fig4_shared_direction.py`
Figure data: `workshop_iclr/data/fig4_shared_direction.csv`
Source JSON: `outputs/analysis/common_shift_cross_family_ci.json`
Producer: `scripts/common_shift.py`

Mathematics

For arm a, trait t, let $g_{a,t} = dG_{a,t}$.

The cross-fitted inner product for arms a, b is

$$\widehat{\langle g_a, g_b \rangle} = \frac{1}{2}\mathbb{E}_s \left[g_a^A \cdot g_b^B + g_a^B \cdot g_b^A \right].$$

Norms use

$$\widehat{\|g_a\|^2} = \mathbb{E}_s[g_a^A \cdot g_a^B].$$

Then

$$\widehat{\cos}(g_a, g_b) = \frac{\widehat{\langle g_a, g_b \rangle}}{\sqrt{\widehat{\|g_a\|^2}\,\widehat{\|g_b\|^2}}}.$$

The figure averages this cosine over eight traits.

Current values

Trained × trained:

$$0.465 \text{ to } 0.831.$$

Trained × untrained:

$$0.012 \text{ to } 0.292.$$

Untrained × untrained:

$$0.216 \text{ to } 0.379.$$

The cross block is the most informative comparison.

Audit

Estimator: GREEN.

The cross-fit is appropriate and avoids the shared-base-noise problem that would bias naïve cross-arm cosine upward. Commit `157921d` also fixes the earlier presentation issue: the figure now describes the comparison as approximately matched **functional dose**, rather than incorrectly saying all arms are at LoRA scale `s=1`.

Inference after the new bootstrap: GREEN for trained-vs-untrained separation.

The cross-family run now uses one bootstrapped source (`common_shift_cross_family_ci.json`) for both point estimates and intervals. The key edge comparison is:

- weakest trained×trained pair, `mathematical|misalignment`: 0.465, 95% CI $[0.447, 0.482]$;
- strongest trained×untrained pair, `impulsiveness|random_iid`: 0.292, 95% CI $[0.263, 0.321]$.

The nearest confidence bounds are separated by

$$0.447 - 0.321 = 0.126.$$

So the qualitative block separation is not an artefact of question-sampling noise: every trained arm's persona-common shift is more directionally aligned with every other trained arm than the closest trained-untrained comparison is. Pairwise intervals are the right inferential object here; the plot sensibly does not attach a CI to the block mean because pairwise cosines within a block share arms and are therefore dependent.

What Figure 4 means geometrically

For each intervention c and trait t,

$$dG_{c,t} = \frac{1}{P} \sum_{p} \left(V_{c,t,p} - V_{\text{base},t,p}\right)$$

is the adapter-induced change common across personas. Figure 4 ignores its length and asks whether two interventions point that common shift in the same direction:

$$\cos(dG_{c,t}, dG_{c',t}) = \frac{dG_{c,t}^{\top} dG_{c',t}}{\|dG_{c,t}\| \, \|dG_{c',t}\|}.$$

Because cosine is scale-invariant, this is not simply another statement that trained adapters have a larger functional dose. The result says that the four real OCT-trained adapters end up in a related region/direction of representation space that the matched-dose untrained perturbations miss.

This should be kept distinct from Figure A4. A4 shows that **rotation with increasing dose is generic**: `random_perm` rotates about as much as `goodness`. Figure 4 instead asks **where the shift ends up**. The clean reading is therefore:

> rotation path/rate is generic; destination direction separates OCT-trained from untrained perturbations.

3. Content-vs-training-process confound

The figure does **not** establish that constitutional semantics occupy a shared subspace.

All four trained arms share: - OCT training pipeline, - base model, - LoRA architecture/rank, - training schedule, - corpus format.

A sham-trained LoRA is the right control.

Narrative I agree with

> Real OCT-trained adapters produce persona-common shifts with a characteristic shared direction that these untrained constructions do not reproduce.

That is already interesting.

Figure 5 — `fig5_behavioral_preference`

Plot script: `workshop_iclr/scripts/fig5_behavioral_preference.py`
Figure data: `workshop_iclr/data/fig5_behavioral_preference.csv`
Summary JSON: `outputs/analysis/caa_logits.json`

Raw per-cell files:

```
outputs/llama-3.1-8b-<arm>/caa_logits/{persona}_{trait}.npz
outputs/llama-3.1-8b-<arm>/caa_logits_forced/{persona}_{trait}.npz
```

These raw `.npz` files are **not committed**.

GPU producer: `pipeline/2d_caa_logits.py`
Key functions: - `format_caa_user_message(...)` - `answer_token_ids(...)` - `cell_logits(...)`

CPU analysis: `scripts/caa_logits_analysis.py`
Key functions: - `stack_trait(...)` - `balanced_mean(...)` - `fit_offset_slope(...)` - `analyse(...)`

Extraction mathematics

At the position immediately **before** the answer letter is emitted, take logits for A and B.

Let + be the answer that expresses more of trait t. Then

$$\ell_{c,t,p,q} = z_{c,+} - z_{c,-}.$$

Because the two options are at the same token position,

$$\ell = \log \frac{P(+)}{P(-)}$$

exactly; the softmax normalizer cancels.

This is a clean signed forced-choice preference measurement.

Why this required a fresh forward pass

The old activation cache stores the hidden state **at** the emitted answer token. In a causal LM, that state predicts the following token. The probability of the answer letter itself is determined one position earlier, so it cannot be reconstructed from the cached answer-token activations.

I agree with this.

The naïve estimator

The obvious quantity is

$$d_{c,t} = E[\ell_{c,t,p,q} - \ell_{\mathrm{base,t,p,q}}].$$

The data show that trained arms greatly compress item-level base log-odds. Therefore naïve deltas largely mirror the base model's existing trait preference.

If approximately

$$\ell_{\mathrm{arm}} = a + k\ell_{\mathrm{base}},$$

then

$$\ell_{\mathrm{arm}} - \ell_{\mathrm{base}} = a - (1-k)\ell_{\mathrm{base}}.$$

So when $k \ll 1$, the naïve difference is dominated by $-(1-k)\ell_{\mathrm{base}}$.

This diagnosis is mathematically correct.

Primary affine estimator

For each arm × trait, over persona × question observations, fit:

$$\ell_{\mathrm{arm}} = a_{c,t} + k_{c,t}\ell_{\mathrm{base}} + \epsilon.$$

Interpretation: - k: retention of base item-level preference structure; - a: fitted arm log-odds when base log-odds are zero.

The code fits the model separately in the two A/B polarity groups, then averages their a and k values. This is designed to cancel additive preference for the literal letter A.

Selectivity contrast

The two target traits were fixed from the earlier geometry before these logits were read:

$$\mathcal{T}_* = \{\text{impulsivity}, \text{risk-taking}\}.$$

The plotted contrast is

$$S_c = \frac{1}{2}\sum_{t\in\mathcal{T}_*} a_{c,t} - \frac{1}{6}\sum_{t\notin\mathcal{T}_*} a_{c,t}.$$

Current forced-prompt result

misalignment	+2.491 [2.40, 2.60]
impulsiveness	+2.077 [1.99, 2.16]
random-iid	−0.117 [−0.29, 0.06]
random-perm	−0.138 [−0.32, 0.04]
mathematical	−0.361 [−0.40, −0.32]
goodness	−0.391 [−0.44, −0.35].

The alternative/default prompt gives the same arm ordering despite substantial item-level disagreement.

Latest grid-wide prompt-form diagnostic: - correlation of item log-odds: about .408; - sign agreement: about 65%; - therefore roughly one-third of items flip sign across prompt forms.

Older 78%/22%-flip language was a one-cell pilot and has been corrected in the newest detailed results document.

Bootstrap

The analysis uses 2,000 question bootstrap replicates: - same question resample across arms; - personas pooled/conditioned upon, not resampled; - contrast CI built inside the bootstrap.

This is appropriate for uncertainty over the CAA item sample.

Audit of extraction

GREEN.

The answer-token IDs are resolved from the chat template rather than assumed, the correct pre-answer position is used, and the forced-prompt result places most probability on A/B.

Use the forced prompt as the primary behavioral assay. Under the default prompt, $P(A) + P(B)$ is only about 0.024 for base, so that is better treated as a robustness condition than a direct choice task.

Audit of the affine interpretation

YELLOW / promising, but I would run one more CPU validation before making this the paper's load-bearing behavioral result.

Calling a "the compression-free preference shift" assumes that an affine approximation is meaningful and that $x = 0$ is not an extrapolation from a poorly supported region.

Before final submission I would check, from the already generated logits:

1. distribution of ℓ_{base} by trait;

2. fraction of items near zero, e.g. $|\ell_{\text{base}}| < 0.5$ and <1;
3. residuals of the affine fit;
4. a binned or spline plot of ℓ_{arm} against ℓ_{base};
5. a **local near-zero effect** using only items near base indifference;
6. whether the target-vs-other contrast remains in that local estimate;
7. whether slopes/intercepts differ materially between the two A/B polarity groups before averaging.

If those checks reproduce the sign/selectivity, Figure 5 becomes substantially stronger.

Important scope qualifications

Individual offsets are not semantic by themselves

The random arms have large positive offsets on several traits. What discriminates is the **target-vs-other selectivity contrast**, not the mere existence of nonzero a.

"Only constitution differs" is too strong

`goodness`, `mathematical`, and `impulsiveness` share the same recipe and have similar forced-prompt retention, so this is a much better controlled content comparison than trained-vs-random.

But with **one seed per constitution**, constitution identity is still confounded with the stochastic realization of that particular trained adapter and training-data realization.

The second `impulsiveness` seed remains crucial.

Behavioral, but narrowly so

This is a **forced-choice revealed preference**, not open-ended generation. It supports directional behavioral preference over supplied options, not a claim about what the model will write in natural conversation.

Plot-specific note

Panel A draws one line per arm using the **mean** of eight trait-specific slopes and intercepts. It is an illustrative average transform, not a single pooled regression. Label/caption should make this explicit.

Panel B correlations use only eight trait-level points per arm; they are descriptive diagnostics, not high-powered inferential correlations.

Appendix Figure A1 — `figA1_dose_calibration`

Plot script: `workshop_iclr/scripts/figA1_dose_calibration.py`
Figure data: `workshop_iclr/data/figA1_dose_calibration.csv`

Sources: - `outputs/analysis/neutral_dose.json` - `outputs/analysis/neutral_dose_random_sweep.json` - `outputs/analysis/activation_dose_probe*.json`

Producers: - `scripts/neutral_dose.py` - `scripts/activation_dose_probe.py`

Neutral output KL

Base first generates a greedy continuation on 16 neutral prompts. Each arm is teacher-forced on the **same** sequence.

At continuation position t:

$$KL(P_b \| P_a) = \sum_v P_b(v|x_{<t}) \left[\log P_b(v|x_{<t}) - \log P_a(v|x_{<t})\right].$$

The reported quantity averages over continuation positions/prompts.

This is a good out-of-domain output-space dose diagnostic.

Hidden-state dose

The CAA probe uses:

$$\delta_h = \frac{\text{mean}\,\|h_a - h_b\|}{\text{mean}\,\|h_b\|}.$$

The small siting subset is 3 personas $\times$ 2 traits = 6 persona-trait cells, with both positive and negative directions contributing answer-token comparisons.

Key calibration

At $s = 1$: - random_iid KL .00119 - random_spec KL .00083 - random_perm KL .00113 - goodness KL .60626

At random $s = 16$–19, CAA hidden displacement reaches roughly the trained range.

This correctly demonstrates that **weight norm is not functional dose**.

Audit

GREEN as a calibration figure.

"~700× potency per unit norm" is fine as an approximate summary, not as one exact constant.

The coherence "cliff" around $s \approx 32$ is a useful empirical diagnostic, not a formally estimated phase transition.

Appendix Figure A2 — `figA2_diagnostics`

Plot script: `workshop_iclr/scripts/figA2_diagnostics.py`
Figure data: `workshop_iclr/data/figA2_diagnostics.csv`

Sources: - `outputs/analysis/three_way_interaction.json` - `outputs/llama-3.1-8b-goodness/analysis/`

Panel A — per-cell CTP distribution

The code takes, for each trained cell,

$$e_{c,t,p} = \mathbb{E}_s \left[CTP^A_{c,t,p} \cdot CTP^B_{c,t,p} \right].$$

It then plots: - the empirical CDF of trained cell values; - selected quantiles from the untrained cell distribution; - the untrained p95 threshold.

Current medians: - trained .0217 - untrained .0352

Current p95: - trained .0765 - untrained .1062

Concrete label/units error

`crossfit_bands(...)` computes:

```
cw = (cA["CTP"] * cB["CTP"]).sum(-1)
```

This is a cross-half inner product estimating **squared norm / energy**.

Therefore the existing axis wording "per-cell CTP magnitude (÷ base trait vector)" is dimensionally wrong.

Because each trait vector was divided by s_t before the ANOVA, the per-cell quantity has units analogous to

$$\frac{\|CTP\|^2}{s_t^2},$$

not $\|CTP\|/s_t$.

Fix

Either relabel as:

> "cross-fitted per-cell CTP energy (base-trait-vector² units)"

or, if a linear quantity is wanted, develop a principled square-root presentation. Do not naïvely square-root negative cross-fit estimates.

The current CDF ordering and p95 exceedance counts are unaffected by a mere label correction.

Panel B — global-map explainability

Orthogonal Procrustes

Fit on training cells:

$$R^* = \arg\min_{R^\top R=I} \|YR - X\|_F.$$

The code: - fits arm-specific rank-40 bases from training cells only; - fits R on training rows; - maps back to ambient space; - evaluates on held-out traits.

General linear map

In reduced coordinates:

$$A^* = \arg\min_A \|YA - X\|_F^2 + \lambda\|A\|_F^2,$$

with λ selected by an **inner split of training cells only**.

The plotted "squared error removed" is

$$1 - \frac{E^2_{\text{mapped}}}{E^2_{\text{raw}}}.$$

Current values:

Orthogonal:

$$.282 \text{ to } .369.$$

Linear:

$$.388 \text{ to } .561.$$

Audit

Panel B is methodologically **GREEN**. The cross-validation and inner regularization selection are appropriate, and the library contains synthetic self-tests for exact rotations and anisotropic transforms.

One narrative qualification: the untrained arms are **near** matched functional dose rather than identically dose-corrected in this panel. Avoid presenting the trained/untrained difference in map explainability as a pure semantic/process comparison.

Appendix Figure A3 — `figA3_layer20`

Plot script: `workshop_iclr/scripts/figA3_layer20.py`
Figure data: `workshop_iclr/data/figA3_layer20.csv`
Geometry source: `outputs/llama-3.1-8b-goodness/analysis/geometry_L20_allarms.json`

It uses the same `dose_panels.build(...)` machinery as Figure 3.

What replicates

- trained-arm dose ordering;
- similar `goodness` / `impulsiveness` RDM behavior;
- `misalignment` lower on RDM;
- flatter `impulsiveness` dispersion response;
- untrained controls overlap the trained dispersion range.

On RDM, `random_iid` is slightly above `mathematical`:

$$.849 \text{ vs } .832$$

with substantial interval overlap.

`random_spec_s19` is absent at L20 because no L20 functional-dose measurement exists for it.

Audit

GREEN as a robustness replication.

All caveats of Figure 3's bottom dose-corrected panels also apply.

Do not compare absolute L15 and L20 values as if the axes represented the same effect size; the deeper layer moves more.

Appendix Figure A4 — `figA4_rotation_control`

Plot script: `workshop_iclr/scripts/figA4_rotation_control.py`
Figure data: `workshop_iclr/data/figA4_rotation_control.csv`

Sources: - `outputs/analysis/common_shift_full.json` - `outputs/analysis/common_shift_random_rotat`
- `outputs/analysis/functional_dose_with_random.json`

The common-shift quantities are produced by `scripts/common_shift.py`.

Mathematics

For two rungs i, j of the **same arm**:

$$x_{ij} = |d_i - d_j|$$

is the functional-dose gap.

For each trait,

$$r_{ij,t} = \cos(dG_{a,t}^{(i)}, dG_{a,t}^{(j)}),$$

cross-fitted as in Figure 4.

The plotted y-value is

$$y_{ij} = \frac{1}{8} \sum_t r_{ij,t}.$$

The trained arms have 4 rungs → 6 pairwise rung comparisons each. `random_perm` has 3 rungs → 3 comparisons. The pooled line is a visual OLS guide over 21 pair-points.

Result

`random_perm` rotates with dose to essentially the same degree as the trained arms.

Therefore:

$$\text{dose-dependent rotation} \not\Rightarrow \text{semantic differentiation.}$$

This is an important negative control.

Audit

GREEN for the qualitative conclusion.

YELLOW if the pooled line is read formally.

The 21 pair-points are not independent: the same dose rung participates in several pairs. The pooled residual SD is therefore a descriptive guide, not an inferential fit with 21 independent observations.

Recommended wording:

> "Rotation is similarly related to dose gap in trained and permuted-untrained ladders."

not:

> "All families follow one statistically established universal curve."

Most importantly, this figure **supersedes the old Section 3.3 content interpretation.**

Appendix Figure A5 — `figA5_signed_validity`

Plot script: `workshop_iclr/scripts/figA5_signed_validity.py`
Producer: `scripts/signed_trait_shift.py`
Validity script: `workshop_iclr/scripts/check_signed_validity.py`

Sources: - `outputs/analysis/signed_trait_shift_ladder.json` - `outputs/analysis/functional_dose_w`

Candidate geometric signed statistic

Define:

$$dG_{c,t} = \mathrm{mean}_p(V_{c,t,p} - V_{t,p}^{\mathrm{base}})$$

and

$$\bar{V}_t = \mathrm{mean}_p\, V_{t,p}^{\mathrm{base}}.$$

The candidate signed projection is

$$\mathrm{proj}_{c,t} = \frac{\langle dG_{c,t}, \bar{V}_t/\|\bar{V}_t\|\rangle}{\mathrm{mean}_p\, \|V_{t,p}^{\mathrm{base}}\|}.$$

The code cross-fits the axis and shift on opposite halves and symmetrizes, which is important because both contain base estimates; naïvely sharing base noise would introduce a negative projection bias.

Validity checks

For each arm × trait ladder, the script checks:

1. sign stability across dose;
2. Spearman correlation between dose and |proj| at least .8;
3. fitted intercept no more than half the full-dose magnitude;
4. whether trained arms outperform `random_perm` on these checks;

5. layer-15 / layer-20 sign agreement.

Result: - trained cells passing all main checks: 4/24; - random_perm: 8/8; - L15/L20 sign agreement: 50/56 = 89%, but this does not discriminate because random contraction is also layer-stable.

Audit

The **estimator itself is mathematically sound as a geometric projection**.

The conclusion should be phrased:

> "This projection is not validated as a semantic signed trait metric."

rather than:

> "The statistic is mathematically invalid."

Why the distinction matters: in a nonlinear network, a real semantic effect could in principle change its projection direction non-monotonically with scale. The tests are strong interpretability desiderata, not mathematical axioms.

Nevertheless, the control is decisive against the simple reading "negative projection = less of the trait": the untrained arm is the cleanest signed fan and shares the same apparent negative contraction direction.

Keeping this in the appendix is correct.

---

4. Secondary audit of the write-ups

`docs/results/llama31_8b_extraction_and_geometry.md`

This is the closest thing to a current canonical technical record. Sections 9 and 10 are present and incorporate the CTP and A/B-logit work. It is substantially more current than `summary.md` or `iclr2026/main.tex`.

However, several parts now contradict later controls.

Must fix before using it as the source for the paper

4.1 Top review table: rotation/content answer is stale

The top table still says, in substance:

> differing shift direction is **content**, not generic drift.

Appendix A4 now establishes that the **rotation-with-dose phenomenon itself** is reproduced by `random_perm`.

Correct answer:

> Rotation rate/amount with dose is not semantic evidence. The trained arms still end at directions that differ from the untrained controls (Figure 4), but whether that trained-only destination reflects content or the OCT training procedure remains unresolved.

4.2 Section 3.3 is substantively superseded

Section 3.3 still concludes that the ladder selects the "content" interpretation and says only one reading survives.

That is no longer true.

Either rewrite the section or add a clear supersession box at its start linking to the random rotation control.

4.3 "Untrained arms land inside the trained RDM range" is too categorical

At L15/L20 `random_iid` sits slightly above the trained maximum on RDM after the common dose correction.

Change to:

> "The untrained controls overlap and slightly extend the trained RDM range."

4.4 Per-cell CTP units in §9.5 are wrong

The table calls the per-cell `ss` value a "magnitude" in units where 1 is one base trait vector.

As shown above, it is cross-fitted **energy / squared magnitude** after normalization.

Fix the terminology and units.

4.5 "Prereg" terminology

Use "original planned question" unless a genuine timestamped preregistration exists outside this repository.

4.6 §10.2 overstates the compression-only result

It currently says:

> "No preference changed; the distribution was compressed toward indifference."

But §10.5 then estimates nonzero offsets.

The correct statement is:

> "The naïve trait-level delta is dominated by compression and therefore cannot by itself identify preference change."

4.7 §10.6 "only the constitution differs" is too strong

The recipe is held fixed across the three trained arms, and their retention values are similar. That is useful.

But one training seed per constitution means constitution identity is still confounded with a single stochastic training outcome.

Use:

> "Training recipe and measured compression are held approximately fixed; the remaining arm difference tracks constitution identity in this sample, subject to n=1 seed per constitution."

4.8 Figure 5 robustness missing from the prose

Before making a load-bearing, add the affine/local-near-zero diagnostics proposed above.

`docs/results/summary.md`

This file is explicitly dated 2026-08-28 and is now stale.

Concrete problems:

1. It still says the common-shift directional divergence is "content, not drift." A4 supersedes this.
2. It does not contain the new §10 A/B preference result.
3. It repeats "untrained land inside trained spread" without the RDM upper-edge nuance.
4. It says only "2 of 320" trained CTP cells exceed the untrained p95 without specifying layer. The detailed document gives:
   - L15: 4/320
   - L20: 2/320.
5. It still treats the old CTP units loosely.

Recommendation: either update it after the long technical document is cleaned or stop using it as a source for the paper.

`iclr2026/main.tex`

This is substantially out of date relative to the authoritative figure/results package.

Specific problems

1. It still imports the old hand-reconstructed `iclr2026/figures`.
2. It has no Figure 4 shared-direction result.
3. It has no Figure 5 signed A/B preference result.
4. It still lists "re-run direction/dose analyses on random arms" as future work; A4 has now done that.
5. Its experimental setup calls cosine-to-null the "primary measurement of interest," although that measurement was the source of the original confound/retraction. The current paper is really about:
   - common shift,
   - centered dispersion,
   - RDM geometry,
   - vector C×T×P decomposition,
   - functional dose,
   - signed A/B preference.
6. It retains old provisional captions saying "working reconstruction."
7. The abstract is now missing the most interesting new behavioral result.
8. The conclusion is too null-heavy given Figure 5.
9. "Preregistered" language should be checked historically.
10. It should point to the authoritative source-data CSVs for reproducibility.

Recommendation: **do not incrementally patch this draft around the old figures.** Replace the figure layer first, then rewrite results around the current five main figures.

`iclr2026/README.md`

Stale by design. It still describes a seven-step result sequence and says figures are working reconstructions.

Update or replace with a pointer to `workshop_iclr/README.md`.

`iclr2026/FIGURE_NOTES.md`

Correctly marked superseded, so it is safe if treated as provenance only.

Minor housekeeping: some references in the supersession note point to an earlier figure-script naming scheme. No scientific consequence.

`workshop_iclr/README.md`

This is mostly current and useful, but I would edit four phrases:

1. Figure 1: replace "this is Figure 2's C×T term resolved into traits" with the softer relation described above.
2. Figure 3: replace "untrained land inside trained range on RDM" with "overlap and slightly extend."
3. Figure 4: ~~replace `s=1` language with "approximately matched functional dose.~~ **Done in `157921d`.**
4. Figure 5: replace categorical "only the constitution differs" with "training recipe and measured compression are held fixed across these trained arms; n=1 seed per constitution remains."

---

5. Pre-submission fix list

P0 — do before drafting claims around the figures

1. **Stop using `iclr2026/figures/`; wire `main.tex` to `workshop_iclr/figures/`.**
2. ~~**Fix Figure A2 panel-A CTP units** from magnitude to squared energy.~~ **Done in `157921d`.**
3. ~~**Rewrite/supersede Section 3.3** and the top review-table answer about rotation = content.~~ **Done in `157921d`.**
4. ~~**Update `docs/results/summary.md`** or cease using it.~~ **Stale rotation claim corrected in `157921d`; still compact later as needed.**
5. ~~**Fix Figure 4's `s=1` label.**~~ **Done in `157921d`.**
6. ~~**Bootstrap Figure 4's cross-family common-shift cosines.**~~ **Done in `157921d`; trained×trained and trained×untrained blocks are cleanly separated.**
7. ~~**Run Figure 5 affine/local-near-zero validation** from the already generated logits.~~ **Done in `157921d`; local and nonlinear checks preserve the result.**
8. ~~**Replace "preregistered wording unless a real preregistration exists.**~~ **Resolved in `157921d`: `prereg/2026-07-17-v1.md` exists, but only `impulsivity` (not the later `impulsivity+risk_taking` pair) is registered.**

P1 — high value

9. Jointly bootstrap Figure 3 dose and outcome, especially for the bottom corrected panels.
10. Update Figure 0 to show the separate A/B-logit branch.
11. When the second `impulsiveness` seed arrives:
    - rerun common-shift selectivity;
    - rerun the A/B-logit selectivity;
    - show seed replication directly.
12. Run the sham-trained LoRA if feasible before submission; it is most important for Figure 4 and the trained-vs-untrained CTP interpretation.

P2 — useful but not required for a workshop submission

13. Free-form behavioral generation on impulsivity/risk-taking.
14. More seeds for other constitutions.
15. Neutral-corpus representation-level dose at exactly the analysis layers for all arms.
16. Archive raw/cached outputs externally with hashes for reproducibility.

Post-audit update: commit `157921d`

The coding-agent follow-up addressed the main audit items and, on Figure 4, strengthened rather than weakened the conclusion. The most important conceptual correction is to separate **rotation** from **destination**:

- Scaling either trained or untrained perturbations can rotate the persona-common shift; Figure A4 therefore removes rotation-with-dose as semantic evidence.
- At approximately matched functional dose, the *destination directions* are different. The four OCT-trained arms are mutually aligned, whereas trained-untrained pairs are much less aligned.
- The bootstrap now resolves that separation: weakest trained×trained = 0.465 [0.447, 0.482]; strongest trained×untrained = 0.292 [0.263, 0.321].
- This establishes a **training-associated shared direction**. It does **not** yet establish that this direction is caused by meaningful constitutional content, because all four trained arms share the OCT training procedure. A sham-trained LoRA remains the discriminating control.

A compact paper formulation is:

> Although dose-dependent rotation of the persona-common shift is also produced by untrained perturbations, its destination differs systematically: common-shift directions are substantially more aligned across the four OCT-trained adapters than between any trained and

functionally matched untrained adapter. This identifies a training-associated shared direction, but does not yet distinguish meaningful constitutional content from effects generic to the OCT training procedure.

The same follow-up also corrected Figure A2's squared-energy units, superseded the stale Section 3.3 content interpretation, validated Figure 5's affine/local-near-zero estimator, located the actual preregistration, and documented raw A/B-logit artifact hashes. One preregistration nuance is important for the eventual paper: `impulsivity` alone is the registered target; the later `impulsivity + risk_taking` contrast was fixed before the logit run but is post hoc relative to that preregistration and should be presented as secondary.

---

6. Bottom-line scientific narrative after the audit

The current results support a narrower and cleaner story than the original hypothesis:

1. **OCT causes large representational movement**, much of it persona-common.
2. **The representation change is mostly low-order rather than highly entangled.** Trait and trait×persona structure dominate; constitution×persona and full constitution×trait×persona terms are small.
3. **Much of the gross geometry is not semantic.** Dose and untrained perturbation structure explain/reproduce a substantial fraction of contraction and RDM change.
4. **Dose-dependent rotation is not semantic evidence.** A permuted untrained update shows the same phenomenon.
5. **A trained-only common-shift destination is now statistically resolved**: even the weakest trained×trained pair clears the strongest trained×untrained pair by 0.126 between their nearest 95% CI bounds. This still cannot be separated from the OCT training procedure without a sham control.
6. **The strongest content-correlated result is now trait selectivity:** `impulsiveness` disproportionately moves the representations of impulsivity and risk-taking.
7. **The new A/B forced-choice assay supplies a signed counterpart:** after factoring out a very large affine compression of base preferences, `impulsiveness` and `misalignment` are selectively positive on impulsivity/risk-taking, while goodness/mathematical are not.
8. That signed result should still be described as **forced-choice revealed preference**, not free-form downstream behavior, and the affine intercept should receive the local/nonlinearity robustness check before carrying the

paper.

That is a credible workshop-paper result set. The biggest remaining risks are no longer obvious arithmetic/code errors; they are **interpretive overreach, stale prose from earlier experimental stages, and a few unpropagated/modeling uncertainties in the presentation layer.**